

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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TASK-1

1. TITLE

Implementation of Basic HTML Elements, Tables, Forms and HTML5 Features

2. AIM

To design and develop basic web pages using HTML by implementing headings, paragraphs, text formatting, lists, hyperlinks, tables, images, forms, semantic HTML5 elements, audio, video and different HTML5 input types.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML document.
- To create headings and paragraphs using HTML.
- To apply different text formatting elements.
- To create ordered and unordered lists.
- To create hyperlinks between webpages and external websites.
- To create tables for displaying structured data.
- To insert images into a webpage.
- To create a student registration form.
- To use text, email and password input fields.
- To use radio buttons and checkboxes.
- To create dropdown lists using the <select> element.
- To use the <textarea> element.
- To understand HTML5 semantic elements.

- To embed audio and video into a webpage.
 - To use HTML5 input types such as email, date, number, color and range.
-

4. THEORY

HTML stands for **HyperText Markup Language**. It is the standard markup language used to create and structure web pages.

An HTML document consists of elements represented using tags. These elements tell the browser how the content should be structured and displayed.

This experiment contains four programs covering different HTML concepts.

Program 1 – Basic HTML Elements

The first program demonstrates basic HTML elements such as:

- Headings
- Paragraphs
- Bold text
- Italic text
- Underlined text
- Subscript
- Superscript
- Ordered lists
- Unordered lists
- Hyperlinks

Program 2 – Tables and Images

The second program demonstrates:

- HTML tables
- Table rows and columns
- Table headings
- Table data

- Images
- Alternative text using the alt attribute

Program 3 – Student Registration Form

The third program demonstrates HTML forms and different form controls such as:

- Text input
- Email input
- Password input
- Radio buttons
- Checkboxes
- Dropdown list
- Textarea
- Submit button
- Reset button

Program 4 – HTML5 Features

The fourth program demonstrates HTML5 features such as:

- Semantic elements
 - Header
 - Navigation
 - Section
 - Article
 - Footer
 - Audio
 - Video
 - HTML5 input types
-

5. ALGORITHM / PROCEDURE

1. Create an HTML file with the .html extension.
 2. Add the `<!DOCTYPE html>` declaration.
 3. Create the `<html>`, `<head>` and `<body>` elements.
 4. Add headings and paragraphs.
 5. Apply text formatting using HTML tags.
 6. Create ordered and unordered lists.
 7. Add hyperlinks using the `<a>` tag.
 8. Create a table using `<table>`, `<tr>`, `<th>` and `<td>`.
 9. Insert images using the `<img>` tag.
 10. Create a student registration form.
 11. Add different input controls to the form.
 12. Add a dropdown menu using `<select>` and `<option>`.
 13. Add a multiline input using `<textarea>`.
 14. Add submit and reset buttons.
 15. Create HTML5 semantic sections using `<header>`, `<nav>`, `<section>`, `<article>` and `<footer>`.
 16. Add audio using the `<audio>` element.
 17. Add video using the `<video>` element.
 18. Implement different HTML5 input types.
 19. Save all HTML files.
 20. Open the files in a web browser.
 21. Verify that all elements are displayed and functioning correctly.
-

6. PROGRAM

basic.html

```
<!DOCTYPE html>
<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>Basic HTML</title>

</head>

<body>

  <!-- Heading and Paragraph -->

  <h1>
```

```
</h1>
```

```
<p>
```

ANITS is one of the leading engineering colleges in Visakhapatnam,
offering quality education in Engineering, Technology, and Management.

```
</p>
```

```
<!-- Text Formatting -->
```

```
<h2>Text Formatting</h2>
```

```
<p>
```

```
<b>Bold Text</b>
```

```
<br>
```

```
<i>Italic Text</i>
```

```
<br>
```

```
<u>Underlined Text</u>
```

```
<br>
```

```
H<sub>2</sub>O
```

```
(Subscript Example)
```

```
<br>
```

```
x<sup>2</sup> + y<sup>2</sup> = z<sup>2</sup>
```

```
(Superscript Example)
```

```
</p>
```

```
<!-- Ordered List -->
```

```
<h2>Courses Offered</h2>
```

```
<ol>
```

```
<li>
```

Computer Science and Engineering

```
</li>
```

```
<li>
```

Artificial Intelligence & Machine Learning

```
</li>
```

```
<li>
```

Electronics and Communication Engineering

```
</li>
```

```
</ol>
```

```
<!-- Unordered List -->

<h2>Facilities</h2>

<ul>

  <li>Library</li>

  <li>Computer Labs</li>

  <li>Sports Complex</li>

</ul>

<!-- Hyperlinks -->

<h2>Useful Links</h2>

<a href="https://anits.edu.in" target="_blank">
  Visit ANITS Website
</a>

<br><br>

<a href="https://www.google.com" target="_blank">
  Visit Google
</a>

</body>

</html>
```

table.html

```
<!DOCTYPE html>
<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>Tables and Images</title>

</head>

<body>

  <h1>Student Details</h1>
```

```
<table border="1" cellpadding="8">
```

```
<tr>
```

```
<th>Roll No</th>
```

```
<th>Name</th>
```

```
<th>Branch</th>
```

```
<th>Email</th>
```

```
</tr>
```

```
<tr>
```

```
<td>A2412655209</td>
```

```
<td></td>
```

```
<td>CSE (AI & ML)</td>
```

```
<td>almas@example.com</td>
```

```
</tr>
```

```
<tr>
```

```
<td>24126552101</td>
```

```
<td>Rahul Kumar</td>
```

```
<td>CSE</td>
```

```
<td>rahul@example.com</td>
```

```
</tr>
```

```
<tr>
```

```
<td>24126552102</td>
```

```
<td>Sneha Reddy</td>
```

```
<td>ECE</td>
```

```
<td>sneha@example.com</td>
```

```
</tr>
```

```
</table>
```

```
<h2>College Logo</h2>
```

```

```

```
<h2>College Campus</h2>
```

```


</body>

</html>
```

form.html

```
<!DOCTYPE html>
<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>Student Registration Form</title>

</head>

<body>

  <h1>Student Registration Form</h1>

  <form>

    <label>Name:</label>
    <br>

    <input type="text" name="name">

    <br><br>

    <label>Email:</label>
    <br>

    <input type="email" name="email">

    <br><br>

    <label>Password:</label>
    <br>

    <input type="password" name="password">

    <br><br>
```

```
<label>Gender:</label>
<br>
```

```
<input
  type="radio"
  name="gender"
  value="Male"
>
Male
```

```
<input
  type="radio"
  name="gender"
  value="Female"
>
Female
```

```
<input
  type="radio"
  name="gender"
  value="Other"
>
Other
```

```
<br><br>
```

```
<label>Hobbies:</label>
<br>
```

```
<input
  type="checkbox"
  name="hobby"
  value="Reading"
>
Reading
```

```
<input
  type="checkbox"
  name="hobby"
  value="Sports"
>
Sports
```

```
<input
  type="checkbox"
  name="hobby"
  value="Music"
>
Music
```

```
<br><br>
```

```
<label>Branch:</label>
<br>

<select name="branch">

  <option>CSE</option>

  <option>CSE (AI & ML)</option>

  <option>ECE</option>

  <option>EEE</option>

  <option>Mechanical</option>

</select>

<br><br>

<label>Comments:</label>
<br>

<textarea
  name="comments"
  rows="4"
  cols="30">
</textarea>

<br><br>

<input
  type="submit"
  value="Submit"
>

<input
  type="reset"
  value="Reset"
>

</form>

</body>

</html>
```

form.html

```
<!DOCTYPE html>
<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>HTML5 Features</title>

</head>

<body>

  <!-- Header -->

  <header>

    <h1>
      ANITS - HTML5 Features Demo
    </h1>

  </header>

  <!-- Navigation -->

  <nav>

    <a href="#">Home</a> |

    <a href="#">About</a> |

    <a href="#">Contact</a>

  </nav>

  <!-- Section -->

  <section>

    <h2>About College</h2>

    <!-- Article -->

    <article>

      <p>
        ANITS is a reputed engineering college that provides
        quality education with modern facilities and advanced technology.
      </p>

    </article>

  </section>

</body>

</html>
```

```
</p>
```

```
</article>
```

```
</section>
```

```
<!-- Audio -->
```

```
<h2>Audio</h2>
```

```
<audio controls>
```

```
<source
  src="media/audio.mp3"
  type="audio/mpeg"
>
```

```

  Your browser does not support
  the audio element.
```

```
</audio>
```

```
<!-- Video -->
```

```
<h2>Video</h2>
```

```
<video width="400" controls>
```

```
<source
  src="media/video.mp4"
  type="video/mp4"
>
```

```

  Your browser does not support
  the video tag.
```

```
</video>
```

```
<!-- HTML5 Input Types -->
```

```
<h2>HTML5 Input Types</h2>
```

```
<form>
```

```
  Email:
```

```
  <input type="email">
```

```
  <br><br>
```

Date:

```
<input type="date">
```

```
<br><br>
```

Number:

```
<input
  type="number"
  min="1"
  max="100"
>
```

```
<br><br>
```

Favorite Color:

```
<input type="color">
```

```
<br><br>
```

Range:

```
<input
  type="range"
  min="0"
  max="100"
>
```

```
<br><br>
```

```
</form>
```

```
<!-- Footer -->
```

```
<footer>
```

```
<p>
  &copy; 2026 ANITS.
  All Rights Reserved.
</p>
```

```
</footer>
```

```
</body>
```

```
</html>
```

7. CODE EXPLANATION

Code	Explanation
<code><!DOCTYPE html></code>	Declares the document as HTML5.
<code><html></code>	Root element of an HTML document.
<code><head></code>	Contains metadata and document information.
<code><title></code>	Specifies the title of the webpage.
<code><body></code>	Contains the visible webpage content.
<code><h1> to <h6></code>	Used to create headings.
<code><p></code>	Creates a paragraph.
<code></code>	Displays text in bold.
<code><i></code>	Displays text in italic.
<code><u></code>	Underlines text.
<code><sub></code>	Displays subscript text.
<code><sup></code>	Displays superscript text.
<code></code>	Creates an ordered list.
<code></code>	Creates an unordered list.
<code></code>	Defines a list item.

<a>	Creates a hyperlink.
<table>	Creates a table.
<tr>	Creates a table row.
<th>	Creates a table heading.
<td>	Creates table data/cell.
	Displays an image.
alt	Provides alternative text for an image.

8. TEST CASES AND EXECUTION DATA

A. basic.html

Test Case	Operation	Expected Result	Actual Result	Status
TC-01	Open basic.html	Basic HTML webpage should display	Displayed correctly	Pass
TC-02	Check text formatting	Bold, italic and underline should display	Displayed correctly	Pass
TC-03	Check lists	Ordered and unordered lists should display	Displayed correctly	Pass
TC-04	Click ANITS link	ANITS website should open	Opened successfully	Pass

B. table.html

Test Case	Operation	Expected Result	Actual Result	Status
TC-01	Open table.html	Student table should display	Displayed successfully	Pass
TC-02	Check images	College images should display	Images displayed	Pass

C. html5.html

Test Case	Operation	Expected Result	Actual Result	Status
TC-01	Open html5.html	HTML5 elements should display	Displayed successfully	Pass
TC-02	Play audio	Audio should play	Audio played	Pass
TC-03	Play video	Video should play	Video played	Pass
TC-04	Use date/number/color/range inputs	HTML5 controls should work	Controls worked	Pass

9. OUTPUT

A. basic.html

Output:

Anil Neerukonda Institute of Technology and Sciences (ANITS)

ANITS is one of the leading engineering colleges in Visakhapatnam, offering quality education in Engineering, Technology, and Management.

Text Formatting

Bold Text

Italic Text

Underlined Text

H₂O (Subscript Example)

x² + y² = z² (Superscript Example)

Courses Offered

1. Computer Science and Engineering
2. Artificial Intelligence & Machine Learning
3. Electronics and Communication Engineering

Facilities

- Library
- Computer Labs
- Sports Complex

Useful Links

[Visit ANITS Website](#)

[Visit Google](#)

B. table.html

Output:

Student Details

Roll No	Name	Branch	Email
24126552098	Mohammed Almas	CSE (AI & ML)	almas@example.com
24126552101	Rahul Kumar	CSE	rahul@example.com
24126552102	Sneha Reddy	ECE	sneha@example.com

College Logo

 ANITS College Logo

College Campus

 ANITS College Campus

C. forms.html

Output:

Student Registration Form

Name:

Email:

Password:

Gender:

☐ Male ☐ Female ☐ Other

Hobbies:

☐ Reading ☐ Sports ☐ Music

Branch:

Comments:

Submit

Reset

D. html5.html

Output:

ANITS - HTML5 Features Demo

[Home](#) | [About](#) | [Contact](#)

About College

ANITS is a reputed engineering college that provides quality education with modern facilities and advanced technology.

Audio



Video



HTML5 Input Types

Email:

Date:

Number:

Favorite Color:

Range:

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10. RESULT

Thus, the basic HTML programs were successfully implemented and executed.

The experiment demonstrated the use of **basic HTML elements, text formatting, lists, hyperlinks, tables, images, forms, form controls, semantic HTML5 elements, audio, video and HTML5 input types.**

All four HTML programs were successfully tested in a web browser and the expected output was obtained.
